

Iterative Prompting

Project: Verified Study Buddy

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Examples and Rubrics

Without opening the old lesson, explain one key idea from Examples and Rubrics. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Sampling and Variability

Without opening the old lesson, explain one key idea from Sampling and Variability. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Predictive and Generative AI

Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Controlled prompt experiments. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Controlled prompt experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Changed independent case

A factual response uses four paragraphs when the contract requires three bullets. Design a controlled prompt revision and evaluation.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A summary has accurate facts but uses unexplained terms. Propose one prompt change and a comparison that checks both accessibility and factual preservation.

6. Solve the given case

A factually correct explanation is too difficult for a nine-year-old.

Make and test one focused revision.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

You change a prompt's audience, length and source rules together; the result improves. What can you conclude, and how would you investigate the cause?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 7 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Maintain prompt version, diagnosed failure, change, and before/after rubric scores.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A summary has accurate facts but uses unexplained terms. Propose one prompt change and a comparison that checks both accessibility and factual preservation.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

13. W2 Trace the worked case

On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

14. W3 Audit the limit

A report omits this qualification: Prompt selection on development cases cannot be reported as independent final performance. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

15. W4 Apply a new case independently

On eight paired cases A passes 1–5; B passes 1–4, 6, 7. Special cases are 5 and 8. Compare overall and special-case results; select by total passes.

My reasoning and evidence

16. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Iterative Prompting

A fictional Study Buddy prompt invents a definition when approved notes omit it.

P asks “answer helpfully”. Revision Q adds “if approved notes lack the answer, say not supplied”.

Hold model settings and four development questions fixed. P passes 1,2; Q passes 1,2,3. Case 4 still fails.

Iterative Prompting

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Locate failure

The target defect is unsupported answers when notes are missing.



2. One controlled change

Q adds the missing-evidence rule; other instructions remain fixed.



3. Paired results

P=2/4; Q=3/4; gain is case 3, with case 4 unresolved.



4. Next action

Inspect case 4 and reserve genuinely fresh questions for final evaluation.

CHANGE ONE THING

Change model settings and note wording at the same time as the prompt.

Explain the changed result using the matching lesson.

What evidence supports improvement here, and what remains unproven?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Maintain a Study Buddy experiment log linking failure, single revision and paired rubric results.

19. Project portfolio milestone

Project: Verified Study Buddy

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

Maintain a Study Buddy experiment log linking failure, single revision and paired rubric results.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Study Buddy prompt invents a definition when approved notes omit it.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Inspect case 4 and reserve genuinely fresh questions for final evaluation.

1. Locate failure
2. Next action
3. One controlled change
4. Paired results

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Next action Inspect case 4 and reserve genuinely fresh questions for final evaluation. An iteration should respond to a recorded failure with a precise change. Comparing the same questions reveals whether other behavior changed too. The invented results show a development gain of one case, not a guarantee that the new instruction always works. Save prompt text, settings, source notes and case-level outcomes.

Practice 2: explain the deciding evidence

Changed-case practice: Change model settings and note wording at the same time as the prompt. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. The comparison no longer isolates the prompt revision.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The comparison no longer isolates the prompt revision. Several changes could explain the altered outputs; record them or run separate controlled comparisons.

Repair a missing building block

Review Examples and Rubrics (ai-3/6). Goal: Write a prompt specification with observable response criteria. Short practice: Explain why neither original nor changed draft is a full pass, and provide a complete repaired card.. Return to this week and explain what you changed.

Review Sampling and Variability (ai-3/4). Goal: Sample from a distribution using explicit cumulative intervals. Short practice: Does $T=0.5$ make B impossible, and does choosing A guarantee factual accuracy?. Return to this week and explain what you changed.

Review Predictive and Generative AI (ai-3/1). Goal: Distinguish discriminative prediction from generation by inspecting the requested output. Short practice: Identify the first unsupported claim and explain what source would be needed before asserting it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
